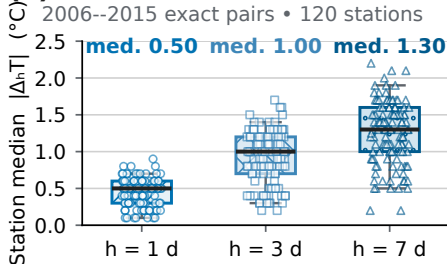


(a) Thermal mismatch



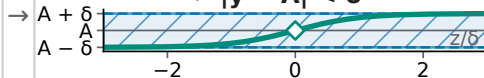
(b) Bounded correction

Canonical form • anchor A • $\delta = 1.0$ °C

$$z = P - A + r\theta$$

$$\hat{y} = A + \delta \tanh(z/\delta)$$

$$\Rightarrow |\hat{y} - A| < \delta$$



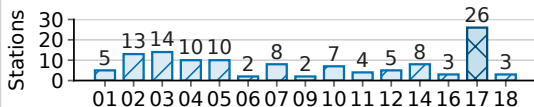
**Deviation bound $\neq$ error bound
 $\neq$ safety guarantee**

(c) Evidence $\rightarrow$ claim boundary

657,480 rows $\rightarrow$ 120 stations $\rightarrow$ 15 pre-attrition

EXCLUDED FROM INPUT

target WTEMP at $t+h$ • horizon-specific future weather



dated inputs
 $\leq t$

$\rightarrow$ exact
keys

$\rightarrow$ station
effects

$\rightarrow$ HUC
sensitivity

$\rightarrow$ claim
gate

HUC2 groups
Frozen gate

✗ ≤ 15 pre-attrition groups
 < 30 required for reporting

✗ $9.54 / 15 = 0.636 < 0.75$

✓ $21.7\% < 25\%$

FIXED-COHORT DESCRIPTIVE ONLY